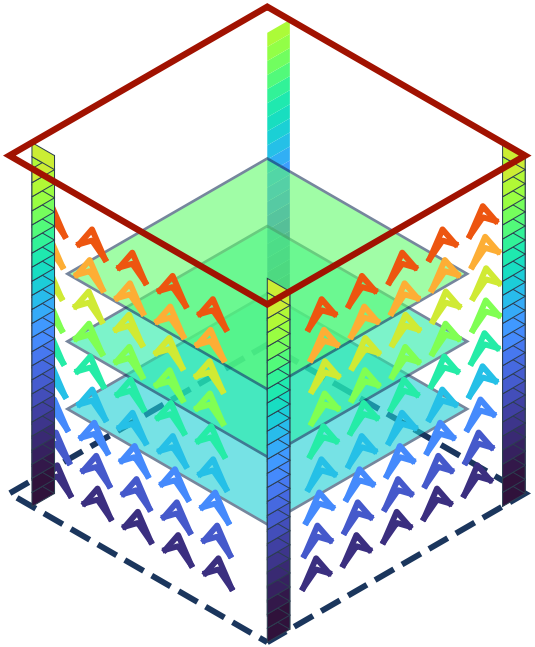
# Finite Element Modal Post-Processing: First Natural Vibration Mode Shape ($f_1 = 579.62$ Hz)

**Mode 1 : Coupled Flexure – Shear**
**$f_1 = 579.62$ Hz** (Clamped Boundary)
**$f_1 = 304.10$ Hz** (Flexible P-POD Boundary)
NASA GEVS Decoupling Limit: $f_1 \geq 100.0$ Hz
Margin of Safety: **$MS_{f1} = +4.80$**



**$\Delta = 0.00$ (Clamped Deployer Rails Interface)**

